

VENTILATING ONTARIO

BY DON FUGLER

Since 1990, mechanical ventilation systems have been mandated in the National Building Code of Canada (NBCC), as new Canadian houses are generally too tight to rely on incidental air leakage to provide sufficient ventilation air. The mechanical ventilation systems described by the NBCC are meant to represent the minimum systems that will ensure adequate ventilation. How adequate that ventilation truly is is a matter of debate—and research.

Code officials in each province generally adopt most of the National Building Code for the provincial editions. However, several provinces have modified the ventilation sections provided by the NBCC. In the case of Ontario, the ventilation sections of the 1993 and 1997 building codes differ from the NBCC and from other provincial codes in several ways.

One of the major changes is that the Ontario codes explicitly recognize exhaust-only ventilation (EOV). Exhaust-only systems sometimes cause problems with combustion venting in new homes, as the level of house depressurization caused by the exhaust system may affect chimney-vented appliances. The Ontario codes address this risk by mandating spillage-resistant appliances and prohibiting wood combustion in houses with EOV systems.

A basic Ontario system consists of a ventilation fan switch (labeled Ventilation Fan) located in the main living area, next to the thermostat. Operation of the switch generally activates a bathroom exhaust fan. Residents are supposed to simultaneously activate the forced-air fur-

nace circulation fan to mix the house's air, evenly distributing the fresh air that is being sucked in through minor holes in the house's envelope. Systems such as this have been shown to work effectively. An exhaust fan working without circulating air does not provide adequate ventilation for all parts of the house. During the code writing process, some commentators suggested that the ventilation fan

national codes and standards adopt the Ontario approach. But the early research and anecdotal data suggested that, while EOV might be popular with builders, homeowners were not using the systems as intended. Canada Mortgage and Housing Corporation (CMHC) decided to conduct a short survey to verify the use and utility of EOV systems.

The research was conducted during the fall of 2003. University students in architecture or engineering surveyed 140 owners of new houses in three parts of Ontario: Ottawa, Toronto, and Guelph. The houses were all built in 1995 or later. The surveyor asked the homeowners what kind of ventilation system they had, how they used the system, when and how often they opened their windows, and whether they had indoor air quality (IAQ) problems. In some houses, they took rudi-

mentary measurements of air flow, using the CMHC garbage bag technique (see "Air Flow Measurements in the Bag," *HE* Sept/Oct '02, p. 8). The data collected on system performance and user satisfaction were largely qualitative.

Most of the homes that were surveyed—76%, or 91 homes—had EOV systems. A principal question for the surveyors was whether homeowners with these systems simultaneously activated the ventilation fan and the furnace circulation fan. They found that most homeowners surveyed were generally unaware that the two fans had to be operated simultaneously. Over 30% of EOV owners never used the systems. Most of the other EOV owners indicated that they used the bathroom fans to control room odor and humidity, but did not use them for general ventilation. Almost 50% of the EOV

TABLE 1. OPERATION OF EXHAUST-ONLY VENTILATION SYSTEMS

Method of Operation	Frequency of Operation Method (n = 91)	Percentage by Method
No idea	9	10%
Only exhaust fan operated	63	69%
Only circulation fan operated	7	8%
Exhaust and circulation fans operated together	12	13%

switch should also turn on the furnace circulation fan, but code writers thought that this would be too complicated or too expensive. Instead, the homeowner would be taught how to run the system and could adapt—code writers thought—to the two-step activation.

In 1994, a study of the new Ontario ventilation systems included 6 houses with EOV in a sample of 24 homes. Most of the houses did not have the required fan switch on the thermostat, and the ventilation switch was not labeled in most houses. The homeowners with EOV did not see their systems as ventilation systems and used them infrequently. Despite these early results, EOV has become the most widely installed system in new Ontario houses. With the advent of changes to the NBCC, and to the ventilation standard, several people recommended that

owners said that their systems were too noisy. A significant number of homeowners identified problems such as stuffiness or window condensation that could be remedied by operating a ventilation system.

Another 16% of the houses had heat recovery ventilators (HRVs); the remaining 8% had other systems. However, the data gathered on the operation of the HRVs and other ventilation systems were less conclusive. There were fewer examples, and the complexity of the installations made it difficult for the surveyors to interpret the results.

The window-opening data were almost incidental to the study, but they are interesting nonetheless. Over 90% of new homeowners in Ontario do open windows; over 40% open windows even during the winter. In midsummer, almost 10% do not open windows at all, which may indicate

continuous use of air conditioning systems. These houses would benefit from midsummer ventilation to provide fresh air to the houses.

The data collected show that EOV—the most common ventilation system in Ontario over the last ten years—is not being used properly. Having separate switches for the designated ventilation fan and the furnace circulation fan means that the two are rarely operated simultaneously. Homeowners do not recognize the bathroom fan as being part of the ventilation system and do not activate it. Yet they experience problems with IAQ and house humidity levels that would benefit from a properly operated ventilation system. Future codes should at least upgrade such systems to interlock the ventilation fan and the furnace circulation fan.



Don Fugler works for the Policy and Research Division of Canada Mortgage and Housing Corporation (CMHC), the federal agency responsible for housing. He manages projects for CMHC in areas such as IAQ, ventilation, combustion spillage, moisture, attics, and contaminated soils.

FOR MORE INFORMATION:

To find out how to conduct the garbage bag test, visit www.cmhc-schl.gc.ca/en/burema/gesein/abhose/abhose_ce46.cfm.